

Biophysics (Bio215)

TA:

Bhumika Choudhary (bhumikac@iiitd.ac.in)

Deeksha Yadav (deekshay@iiitd.ac.in)

Dr. Arjun Ray
(arjun@iiitd.ac.in)

Google classroom

Quick Links

+ Add link

 Meet

⋮

Generate link

Class code

⋮

Icebo5f7 

 New announcement

 Repost

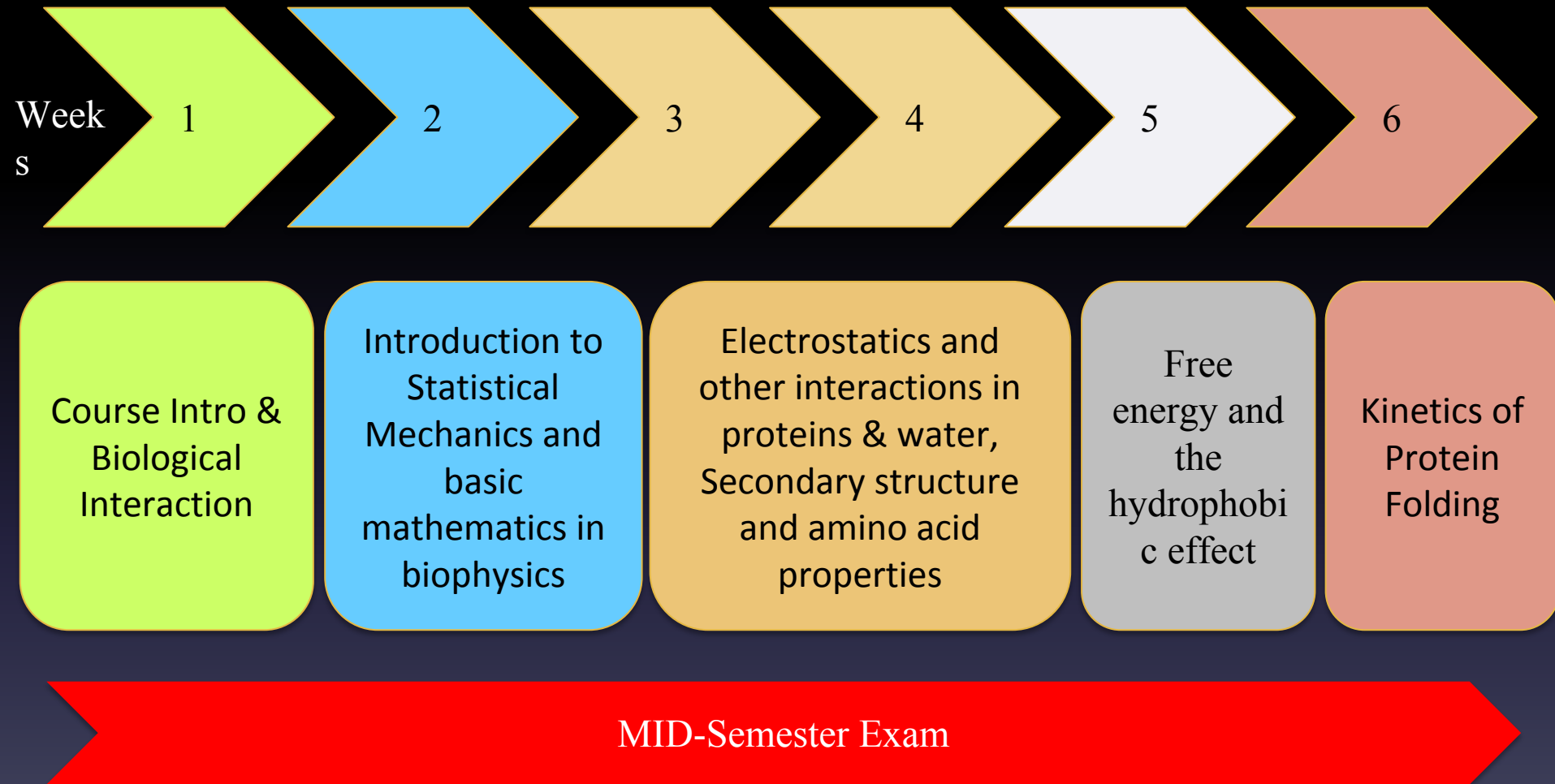


This is where you can talk to your class

Use the stream to share announcements, post assignments, and respond to student questions

 Stream settings

Course Layout



External

Course Assessment

Mid-Semester Exam (Online/Viva)	20
End-Semester Exam (Online/Viva)	30
Quizzes (2/3)	20
Assignment	20
Attendance/Interaction	10

Minimum marks to pass the course: 35

Attendance (Grading scheme)

$$100-80 \% = 10$$

$$79-60 = 9$$

$$59-40 = 8$$

$$39-20 = 7$$

$$< 20 = 1$$

Course literature

We will cover most of the topics here, but it's important that you read literature / papers too

Important part of your training as a scientist - the second you have your degree, most of your learning will come from reading

Main course literature (with reading instructions):

- “Protein physics”, Alexei Finkelstein & Oleg Ptitsyn, ISBN 0-12-256781-1, Can be a bit hard to get, try online. Second version now available, also online as eBook.
- “Physical Biology of the Cell”, Rob Phillips, Jane Kondev, Julie Theriot and Hernan G. Garcia, ISBN 978-0-8153-4450-6. 2nd Ed.

	Misconduct/use of unfair means in assignments/projects	Penalty
1.	First misconduct/use of unfair means during the entire stay in IIIT-Delhi	Zero in the assignment (awarded by faculty) + one letter grade less in the course.
2.	Second misconduct/use of unfair means during entire stay in IIIT-Delhi	Student is assigned an F grade in the course.
3.	Third or further misconduct/use of unfair means during the entire stay in IIIT-Delhi	Student is assigned an F grade in the course and the case is reported to DAC, who may suspend the student for 1 semester to a year.

	Misconduct/use of unfair means in quiz/midsem/endsem	Penalty
1.	First misconduct/use of unfair means during the entire stay in IIIT-Delhi	Student is assigned an F grade in the course.
2.	Second misconduct/use of unfair means during the entire stay in IIIT-Delhi	Student is assigned an F grade in the course and student may be suspended from the program for 1-2 semesters by DAC.
3.	Third misconduct/use of unfair means during the entire stay in IIIT-Delhi	Student is assigned an F grade in the course and student's program may be terminated by DAC.

Ray Lab

- Structure prediction
- Ligand/Drug – Protein interaction
- Molecular dynamics simulation
- Other biological structural problems (DNA, RNA, miRNA, lncRNA etc.)

Biology

Chemistry

Physics

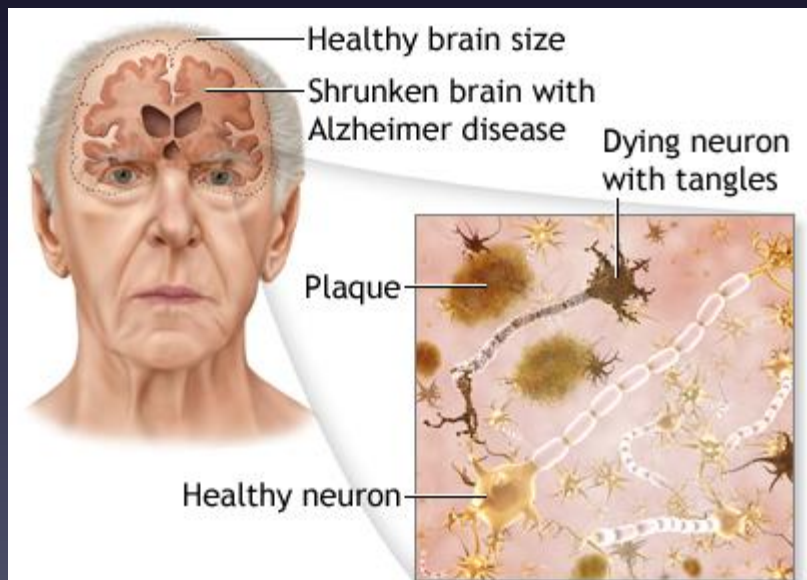
Computer Science

What is biophysics about?

- Understanding nature from simple principles
- Explaining complex process from atoms
- Understanding macromolecular structure
- Understanding measurements & fluctuations
- “Known unknowns & unknown unknowns”
- Prediction: Spectra, measurements, function
- The power of models: You should always simplify as much as possible, but never more
- Understanding WHY, not just observe Modern computer models - simulations

Biophysics applied to proteins

- Understanding protein structure
- Understanding protein folding
- Explaining protein stability
- Sequence-structure-function
- Why do proteins look the way they do?
- Predicting protein structure
- Engineering protein structure
- Mix of chemistry/biology/physics



ADAM.



What is to be gained?

- Quantification of natural biological phenomena
- Two/Three BTP positions in Ray-lab